

Project Technical Report

v11 — Full Feature Development Session

Generated June 20, 2026

Project	AvatarFit — AI-powered personal form coach
Tech Stack	MediaPipe Pose · HTML5 Canvas · Web Speech API · Vanilla JS
Exercises	Squat · Bicep Curl · Lunge · Plank
Key Files	workout.html · workout.js · skeleton.js · voice.js · avatars.js
Report Type	Full development session — architecture, features, bugs & fixes

1. Project Overview

AvatarFit is a browser-based fitness coach that uses your webcam and Google's MediaPipe Pose model to track your body in real time, count repetitions, score your form, and give spoken coaching tips via the Web Speech API. Instead of rendering the raw camera feed, the app draws a stylised vector-capsule avatar on an HTML5 Canvas, coloured green for good form and red when errors are detected.

The project supports four exercises — squats, bicep curls, lunges, and plank — plus a Workout Mode that chains all four in sequence with rest screens, exercise intros, and a comprehensive final summary. A main menu (index.html) lets the user pick their avatar and exercise before launching the workout session (workout.html).

1.1 Architecture

The codebase is intentionally kept framework-free — pure HTML, CSS, and JavaScript:

<code>workout.html</code>	Main app shell. All overlays (summary, workout, rest, intro, gesture indicators) live as hidden divs.
<code>workout.js</code>	Core logic: session state machine, rep counting, form checking, HUD updates, workout flow, gestures.
<code>skeleton.js</code>	Converts raw MediaPipe landmark arrays to named pixel-space joint objects with extrapolation for off-screen joints.
<code>voice.js</code>	Speech synthesis (<code>speak()</code>) with per-text cooldowns; speech recognition for voice commands.
<code>avatars.js</code>	Vector capsule avatar renderer for characters 1–3.
<code>girl_avatar.js</code>	PNG sprite-based renderer for character 4 (Zoe).
<code>themes.js</code>	Canvas background themes and visual effects.
<code>index.html</code>	Main menu: character selection, exercise picker, and Full Workout launch button.

1.2 Session State Machine

All session logic gates on a `sessionPhase` variable with these states:

- **waiting_permission** — camera not yet granted; permission overlay shown
- **waiting_frame** — camera active but user not yet fully in frame
- **countdown** — 3-2-1 countdown before reps begin
- **active** — live tracking; reps counted, form checked every frame
- **paused** — session paused (stop button)
- **ended** — summary displayed; tracking stopped

2. Features Implemented This Session

2.1 Per-Exercise Form Checking

Each exercise has a dedicated branch in `checkForm(joints, ex)`. The function returns `{ok, formOk, ang, formError}` where `ok` drives rep-counting thresholds and `formOk` drives the avatar colour and feedback banner. When `formOk` is false, the specific `formError` string is passed through `updateHUD()` → `showTip()` → `speak()` for coaching.

Squat	Knee valgus (cave-in ratio > 0.18), forward lean (extreme hunch, nose near shoulder line, threshold $sw \times 0.15$)
Lunge	Knee valgus at relaxed threshold (> 0.26 — slightly looser than squat after tuning)
Bicep Curl	Elbow flare (elbow drifts > 30% of shoulder-width laterally during the curl)
Plank	Hip sag (hip drops below shoulder–ankle line) and hip pike (hips too high)

2.2 Squat & Lunge Depth Detection (Post-Rep)

Rather than checking depth in real time (which caused false positives mid-descent), depth is tracked via `minDescentAngle` — the lowest knee angle seen during a descent. The check fires after the rep completes:

- `hipInDescent` flag is set when hip Y crosses `downThreshold` (`hipBaseline` + 7% of canvas height).
- `minDescentAngle` tracks the minimum knee angle throughout the descent.
- On hip rise above `upThreshold`, `countRep()` fires. If `minDescentAngle` > 110°, a shallow-rep tip is shown 900 ms later.
- The depth tip bypasses `showTip()`'s global 3-second debounce by writing directly to `tipEl`, avoiding blocking real-time form tips.

This design means the depth warning 'Go deeper — aim for thighs parallel to the floor' always fires after the rep without interfering with the real-time knee-valgus coaching pipeline.

2.3 Too-Fast Rep Detection

`lastRepTime` is compared on each `countRep()` call. If the interval is below the per-exercise minimum (squat: 2 s, curl: 1.5 s, lunge: 2.5 s), the rep is flagged as `tooFast`. The tip fires and — crucially — `errorTypeCounts['tooFast']` is incremented so the mistake appears in the session summary under 'Focus for next time'.

2.4 Workout Mode

Workout Mode chains all four exercises in a fixed plan: 12 squats → 12 curls → 10 lunges → 30 s plank. The `WORKOUT_PLAN` constant defines each step's exercise key and target. The flow is:

- User clicks 'Full Workout' on the main menu or the ■■ button in-app.
- `startWorkoutMode()` resets all counters and calls `showExIntro()` for the first exercise.
- Each exercise intro overlay shows the goal, a tip, and prompts 'Do one rep to start' (voice line plays). The intro dismisses automatically on the first rep.
- On reaching the target (reps or hold time), `completeWorkoutStep()` fires, saves a `workoutLog` entry, and shows either the rest screen or final summary.
- Rest screens show the completed exercise stats and a 15-second countdown to the next exercise, with a 'Skip Rest' button.
- `setExercise()` guards prevent manual exercise-switching during workout mode.

Each `workoutLog` entry captures: exercise key, `form%` (`goodFrames/totalFrames`), `errorCount`, `errorTypeCounts` snapshot, duration, `reps/holdMs`.

2.5 Hand-Raise Gestures

The `checkStopGesture()` function runs every frame during active sessions. Two gestures are recognised — both require 3 seconds of sustained hold:

- **Right hand raised** (wrist above shoulder, to the right of body centre) → End session / show summary. Pink ring fills on the top-right indicator.
- **Left hand raised** (wrist above shoulder, to the left of body centre) → Cycle to next exercise (squat→curl→lunge→plank→squat). Green ring fills on the top-left indicator.

If both hands are raised simultaneously, the stop gesture takes priority and the switch gesture is cancelled. Each has its own timestamp (`gestureStopStart` / `gestureSwitchStart`). The timer was reduced from 4 seconds to 3 seconds based on user feedback.

2.6 In-App Avatar Picker

A 'Change' button in the sidebar 'Your Avatar' panel opens a full-screen modal (`#avatarModal`) showing the same four character portrait cards (inline SVG) as the main menu. Clicking a card calls `setCharacter(num)` which updates `currentChar`, syncs both the sidebar thumbnails and modal cards' active states, speaks the name, and closes the modal. The avatar updates on the next canvas render frame.

2.7 Main Menu — Workout Mode Button

The `index.html` main menu gained a full-width gold 'Full Workout' card below the 4-exercise grid. It displays the plan ('12 Squats · 12 Curls · 10 Lunges · 30s Plank') and navigates to `workout.html?char={selectedChar}&workout=true`, launching directly into Workout Mode with the selected avatar.

3. Session Summary Screens

3.1 Single-Exercise Summary

Shown when the user ends a free-mode session. Displays: elapsed time, total reps (or plank hold time), form score %, and a 'Focus for next time' section. This section iterates `errorTypeCounts` sorted by frequency (top 3) and shows each error's label, occurrence count, and a specific fix tip from `ERROR_DISPLAY`.

3.2 Workout Summary

Shown after all four exercises complete. The summary was significantly expanded during this session:

- **Total Time** — full workout duration
- **Total Reps** — sum of reps across all rep-based exercises
- **Avg Form %** — mean of per-exercise form scores
- **Mistakes** — total error events across the whole workout

Each per-exercise row now expands to list that exercise's specific errors inline (e.g. '3× Knees caving in', '1× Reps too fast').

The 'All Mistakes' section at the bottom shows every error type with no cap (previously limited to 3), sorted by frequency, with the section title showing the total mistake count: 'All Mistakes (N total)'.

4. Bugs Encountered & Fixes

This section documents every significant bug found during development and how it was resolved. Many issues arose from subtle interactions between the real-time rendering pipeline, tip debouncing, and the rep-counting state machine.

Bug 1 — Red legs on perfect reps

[FORM FALSE
POSITIVE]

Problem

The real-time depth check inside `checkForm()` compared the current knee angle against 110° and set `formOk = false` when the angle was large. However, at the very START of every squat descent, the knee angle is naturally large (the person is just beginning to bend). This caused the avatar legs to flash red on every single rep — even perfect ones.

Fix

Removed the real-time depth check from `checkForm()` entirely. Depth detection was moved fully into the post-rep branch of `updateReps()`, using the `minDescentAngle` tracker. The avatar now only turns red for valgus or forward lean, not for normal descent.

Bug 2 — Knee valgus voice blocked after shallow-rep tip

[DEBOUNCE
COLLISION]

Problem

After a shallow rep, the depth tip was shown using `showTip('shallowSquat', force=true)`. The `force` flag bypassed the 3-second global debounce check, but still set `lastTipTime`. This meant that any knee-valgus tip fired in the following 3 seconds was silently dropped — the user got no voice feedback for real-time valgus errors.

Fix

Changed the depth post-rep tip to bypass `showTip()` entirely. It writes directly to `tipEl.textContent` and calls `speak()` independently, without touching `lastTipTime`. The real-time valgus pipeline now operates on its own debounce cycle, unaffected.

Bug 3 — `startCountdown` vs `runCountdown` naming

[REFERENCE
ERROR]

Problem

Early Workout Mode code called `startCountdown(3, callback)` in several places. The actual function in the codebase is named `runCountdown(onDone)` with no count parameter.

Fix

All occurrences replaced with `runCountdown(callback)`.

Bug 4 — tooFast errors missing from summary

[MISSING
TRACKING]

Problem

The `tooFast` flag in `countRep()` correctly showed the real-time tip via `showTip('tooFast')`, but it never incremented `errorTypeCounts['tooFast']`. As a result, too-fast reps were coached in real time but completely invisible in the end-of-session summary.

Fix

Added `errorTypeCounts['tooFast'] = (errorTypeCounts['tooFast'] || 0) + 1` immediately before the `showTip` call in `countRep()`.

Bug 5 — Workout Mode double-fire on completeWorkoutStep

[RACE
CONDITION]

Problem

If the workout auto-complete trigger and a manual stop gesture both fired within the same frame, `completeWorkoutStep()` could be called twice, causing duplicate rest screens or double-saves to `workoutLog`.

Fix

Added a guard at the top of `completeWorkoutStep()`: if `(sessionPhase === 'ended')` return. Since the function sets `sessionPhase = 'ended'` on first call, any subsequent call within the same session is silently dropped.

Bug 6 — Forward lean detection too sensitive

[THRESHOLD
TUNING]

Problem

The initial forward lean check used a threshold of $sw * 0.55$ (nose within 55% of shoulder-width above the shoulder midpoint). This triggered during normal squats when the user simply leaned slightly forward — a common and acceptable squat stance.

Fix

Threshold tightened to $sw * 0.15$. The nose must now be within 15% of shoulder-width above the shoulder line before the error fires. This only catches genuinely extreme hunching where the chin approaches shoulder height. The user can adjust this constant if desired.

Bug 7 — Lunge valgus threshold too strict

[THRESHOLD
TUNING]

Problem

The knee-valgus check used the same 0.18 threshold for both squats and lunges. In a lunge, the front knee naturally tracks slightly inward relative to the ankle due to the staggered stance, causing frequent false positives for users with normal lunge form.

Fix

Split the threshold: squats remain at 0.18, lunges raised to 0.26. The lunge valgus flag now only fires when the knee caves in noticeably beyond normal variance.

5. Key Design Decisions

5.1 No Framework — Pure Vanilla JS

The entire app runs without React, Vue, or any build step. All state lives in module-level variables in `workout.js`. This makes the app instantly loadable from the filesystem (`file://` protocol) without a server, which is essential since MediaPipe requires camera access from a context that may not always have a local server.

5.2 Post-Rep vs Real-Time Form Checks

For errors that are ambiguous mid-rep (depth, speed), checks are deferred to the post-rep moment. Real-time checks (valgus, forward lean, elbow flare, hip sag) fire every frame but are gated behind `BAD_FORM_CONFIRM_FRAMES = 18` consecutive bad frames (~0.6 s at 30 fps) before triggering audio and error counts. This eliminates noise from brief transitional positions.

5.3 Tip Debouncing Strategy

Two independent debounce layers prevent audio spam: (1) `showTip()` has a 3-second global cooldown via `lastTipTime` — no tip of any kind fires within 3 s of the last. (2) `speak()` in `voice.js` has a 5-second per-text cooldown — the same sentence cannot be spoken twice within 5 s. Post-rep depth tips bypass layer (1) by writing directly to the DOM and calling `speak()` without touching `lastTipTime`.

5.4 Avatar System

Characters 1–3 (Alex, Max, Sam) use a procedural vector-capsule renderer in `avatars.js` that draws body segments as rounded rectangles, scaling and rotating them based on joint positions from `skeleton.js`. Character 4 (Zoe) uses a PNG sprite sheet (`girl_avatar.js`) with individual limb images composited at runtime. Both renderers accept the same (`ctx`, `joints`, `formOk`, `ex`, `charNum`) interface so `workout.js` does not need to know which renderer is active.

5.5 'Do One Rep to Start' — No Button Required

Rather than requiring a button click to begin each exercise in Workout Mode, the intro overlay stays up until the first rep is counted. The voice line 'Do one rep to start' is played immediately when the intro appears, and `countRep()` calls `hideExIntro()` on `repCount === 1`. Detection runs the entire time the intro is shown, so there is zero latency between the first rep and the overlay dismissing.

6. File Change Summary

workout.js

- Added Workout Mode: `WORKOUT_PLAN`, `workoutLog`, `workoutStepIdx`, `workoutStartTime`, `workoutPlankDone` state vars
- Added `startWorkoutMode()`, `completeWorkoutStep()`, `showRestScreen()`, `advanceWorkoutStep()`, `saveWorkoutStep()`, `exitWorkoutMode()`, `showWorkoutSummary()`
- Added `showExIntro()` / `hideExIntro()` — exercise intro overlay with first-rep auto-dismiss
- Added depth tracking: `hipInDescent` flag, `minDescentAngle`, post-rep tip in `updateReps()`
- Added `errorTypeCounts` accumulator — incremented for all error types including `tooFast`
- Added `checkStopGesture()` — right hand stop (3s), left hand switch exercise (3s), separate timers & indicators
- Added `cycleExercise()` — cycles squat→curl→lunge→plank→squat on left-hand gesture
- Added `setCharacter(num)` and avatar modal JS (open/close/select handlers)
- Modified `checkForm()` squat/lunge branch: valgus threshold split (0.18 squat, 0.26 lunge), forward lean detection added
- Modified `showSummary()` to redirect to workout summary when `workoutMode` is true
- Modified `showWorkoutSummary()` to show total reps, per-exercise error breakdown, all mistakes (no cap)
- Modified `countRep()` to increment `errorTypeCounts['tooFast']` when reps are too fast
- Fixed red-legs false positive by removing real-time depth check from `checkForm()`
- Fixed valgus-voice blocked by depth tip — depth tip no longer touches `lastTipTime`

workout.html

- Added `#workoutBtn` in topbar (gold gradient)
- Added `#workoutProgressBadge` (current step indicator)
- Added `#workoutExIntroOverlay` with `.wxi-*` styles
- Added `#restOverlay` with rest countdown and next-exercise preview
- Added `#workoutSummaryOverlay` with total-time, total-reps, avg-form, mistakes stats, per-exercise rows with error breakdown, all-mistakes section
- Added `#gestureIndicator` (right, pink, stop) and `#gestureSwitchIndicator` (left, green, switch) with SVG ring animations
- Added `#avatarModal` with full inline SVG portrait cards for all 4 characters
- Added 'Change' button to sidebar 'Your Avatar' panel
- Updated `.wsum-ex-row` to support nested `.wsum-ex-top` + `.wsum-ex-errs` for per-exercise mistake breakdown
- Added `.wsum-section-label` and `.wsum-ex-err-item` styles
- Bumped `workout.js` script tag from `v=19` to `v=21`

index.html

- Added `.btn-full-workout` CSS (gold gradient, same sheen as exercise cards)
- Added Full Workout button HTML inside `.exercise-section` (below exercise grid)
- Added `workoutModeBtn` click handler navigating to `workout.html?char=N&workout;=true`

7. Known Limitations & Future Work

7.1 Forward Lean Detection

The current forward lean check is a front-camera approximation: it measures how close the nose is to the shoulder midpoint in 2D screen space. True forward lean (torso tilting toward the camera) is a sagittal-plane motion that is not directly observable from a frontal view. The check is most effective for severe hunching where the chin drops toward the chest. A depth camera or side-view camera would enable accurate torso-angle measurement.

7.2 Lunge Leg Side

The lunge currently measures the angle on the left leg only (computeAngle uses leftHip → leftKnee → leftAnkle). Users who lead with the right leg may receive inaccurate rep counts. A future improvement would detect which leg is forward and dynamically pick the appropriate joint chain.

7.3 No Persistent Storage

Session data is not saved between visits. All errorTypeCounts, reps, and form scores are lost when the page closes. A future version could use localStorage or IndexedDB to persist workout history and show progress over time.

7.4 Voice Recognition Conflicts

The Web Speech API's recognition engine calls speechSynthesis.cancel() before each speak() call. If a long coaching tip is playing and a rep fires a 'Rep N!' priority call, the tip is cut off. Priority speech (rep counts) deliberately interrupts non-priority tips, which is generally the desired behaviour but can feel abrupt during dense error periods.

7.5 MediaPipe Landmark Extrapolation

When knees or ankles fall outside the camera frame, skeleton.js extrapolates their positions using the torso direction vector and estimated limb proportions. Extrapolated joints are less accurate and may cause brief false-positive valgus or depth readings for users who stand very close to the camera.

8. Key Constants Reference

Constant	Value	Description
<code>BAD_FORM_CONFIRM_FRAMES</code>	18	Consecutive bad frames before error is counted (~0.6 s at 30 fps)
<code>HIP_DOWN_RATIO</code>	0.07	Hip must drop > 7% of canvas height below baseline to enter descent
<code>MIN_REP_MS (squat)</code>	2000 ms	Minimum time between squat reps before 'too fast' fires
<code>MIN_REP_MS (curl)</code>	1500 ms	Minimum time between curl reps
<code>MIN_REP_MS (lunge)</code>	2500 ms	Minimum time between lunge reps
<code>Valgus threshold (squat)</code>	0.18	Max allowed (knee.x - ankle.x) / shoulderWidth before flagging
<code>Valgus threshold (lunge)</code>	0.26	Same check, relaxed for lunge stance geometry
<code>Depth threshold</code>	110 deg	Minimum knee angle during descent; shallower = shallow rep warning
<code>Forward lean threshold</code>	sw * 0.15	Nose must be within 15% of shoulder-width above shoulder line
<code>Gesture hold time</code>	3000 ms	Duration right/left hand must be raised to trigger stop/switch
<code>Tip debounce (global)</code>	3000 ms	Minimum gap between any two coaching tips (showTip lastTipTime)
<code>Speak cooldown</code>	5000 ms	Per-text cooldown in voice.js for non-priority speech
<code>Speak cooldown (priority)</code>	500 ms	Per-text cooldown for priority speech (rep counts)
<code>WORKOUT_PLAN</code>	4 steps	squat×12, curl×12, lunge×10, plank×30 s
<code>REST_DURATION</code>	15 s	Rest countdown between workout exercises